

Hisu Kim

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EDUCATION

Gwangju Institute of Science and Technology, South Korea <i>M.S. in Climate Analysis and Modeling Lab (Advisor: Jin-ho Yoon)</i>	Sep. 2025 – Aug. 2027 (expected) TGPA 4.37/4.5
Gwangju Institute of Science and Technology, South Korea <i>B.S. in Electrical Engineering and Computer Science and Earth Science and Environmental Engineering Minor in Mathematics</i>	2019 – 2025 TGPA 4.10/4.5
Warsaw University of Technology, Poland <i>Exchange student, Faculty of Building Services, Hydro and Environmental Engineering</i>	Oct. 2022 – Feb. 2023 TGPA 4.86/5.0
University of California, Berkeley, USA <i>Summer Session C</i>	June – Aug. 2023 TGPA 4.0/4.0

PUBLICATIONS AND CONFERENCES

Publications

- **Kim, H.**, Ryu, J., Son, S., Jeong, J., Kim, H., & Yoon, J.-H. (2026). A spectral test of the butterfly effect and physical consistency in the diffusion-based GenCast’s ensembles. *npj Climate and Atmospheric Science*. <https://doi.org/10.1038/s41612-026-01380-1>
- Ryu, J., **Kim, H.**, Wang, S.-Y., & Yoon, J.-H. (2026). Increasing resolution and accuracy in sub-seasonal forecasting through 3D U-Net: The western US. *Geoscientific Model Development* 19(1), 27–39. <https://doi.org/10.5194/gmd-19-27-2026>

Conferences

- **Kim, H.**, Yoon, J.-H., & Ryu, J. (2026, December 7–11). What’s Going on Inside GraphCast? Diagnosing GraphCast’s Latent Graph Signals Behind Blurry Forecast. *Submitted at AGU26 Annual Meeting*, San Francisco, CA, United States.
- **Kim, H.**, Ryu, J., & Yoon, J.-H. (2026, October 11–14). Tracing Blurry Forecast Inside GraphCast: A Latent Space Diagnosis via Spectral Graph Theory. *Submitted to the Asian Conference on Meteorology 2026*, Gyeongju, South Korea.
- **Kim, H.**, Ryu, J., & Yoon, J.-H. (2026, October 11–14). Spectral Diagnosis of GenCast’s Jet Stream Kinetic Energy: Noise Artifacts in Probabilistic AI Weather Models. *Submitted to the Asian Conference on Meteorology 2026*, Gyeongju, South Korea.
- **Kim, H.**, Ryu, J., Jang, W., Kwon, M., & Yoon, J. (2026, September). Uncertainty Assessment in Ensemble Forecast Post-processing Using U-Net and Diffusion. *Accepted at S2S2D 2026*, Reading, United Kingdom
- **Kim, H.**, & Yoon, J.-H. (2025, December 16). Revealing the butterfly effect in deep learning weather prediction model via kinetic energy spectra change. *Oral presentation at AGU25 Meeting*, New Orleans, LA, United States.

PROFESSIONAL DEVELOPMENT

Rosshypalooza 2026 <i>University of Chicago, IL, USA</i>	20 – 31 July 2026
• Project lead: Prof. Pedram Hassanzadeh & Prof. Y. Qiang Sun • Presented “40 vs 10³⁰ years? How much data is needed to predict the weather?”	
Advanced School on Foundation Models for Scientific Discovery <i>ITCP, Italy (virtual)</i>	10 – 18 July 2025
Summer School of Mathematics: Algorithms <i>Eötvös Loránd University, Hungary</i>	23 – 27 June 2025

SCHOLARSHIPS AND AWARDS

2026 Weather Big Data Contest <i>Korea Meteorological Administration</i>	Aug. 2026
• Grand Prize, Minister of Climate, Energy and Environment Award	
Presidential Science Scholarship for Graduate Student <i>Ministry of Science and ICT</i>	2026-7 (1-year)
Government Funded Scholarship for GIST Student <i>GIST</i>	2019 Spring – Present
Scholarship for Outbound Program <i>GIST</i>	2023 Fall
Academic Excellence Scholarship <i>GIST</i>	4 Semesters
Scholarship for Summer Session Program Abroad <i>GIST</i>	2023 Summer

TEACHING EXPERIENCE

Calculus (Single- & Multivariable) <i>Teaching Assistant</i>	2025 Spring – 2026 Fall
• Single-variable (2026 Spring): designed weekly problem sets, prepared and led recitation sessions, and provided written feedback on students' solutions • Multivariable (2025 Spring & Fall, 2026 Fall): held weekly office hours, answering conceptual questions and guiding students through theorem proofs	
Graph Theory <i>Teaching Assistant</i>	2024 Fall
• Conducted weekly office hours, answering questions and guiding students through theorem proofs • Organized regular sessions to provide detailed feedback on homework and quizzes, and advanced mathematical $\text{\LaTeX}$ usage • Offered supplementary lessons on challenging topics and extra problems	
Literature Courses from Prof. Soo-Jeong Lee <i>Teaching Assistant</i>	2023 Spring – 2026 Spring
• Built syllabus generator website to streamline the professor's course preparation process • Used $\text{\LaTeX}$ to edit e-books compiled from students' story-writing assignments	
Digital Design <i>Teaching Assistant</i>	2023 Spring
• Conducted office hours twice a week answering questions from homework • Created sample exam questions to facilitate student learning	

WORK EXPERIENCE

Cybersecurity Analyst	Feb. 2021 – Oct. 2022
<i>Republic of Korea Naval Fleet Command</i>	<i>Computer Emergency Response Team</i>
• Co-administered dozens of Linux production servers across the Fleet Command network • Mitigated <i>log4j</i> and <i>MS-MSDT</i> vulnerabilities via NAC-distributed registry patches and removal of exploitable files; analyzed firewall/IPS packets and logs to flag unauthorized or anomalous communications	

SKILLS

Programming	Python	(Main)
	C/C++, C#, JavaScript, Bash	(Intermediate)
Libraries	PyTorch (Lightning), JAX, xarray, Dask, zarr	
Tools	Linux, Git, $\text{\LaTeX}$, NCL, GrADS	
Language	Korean (native)	
	English (fluent)	
	Spanish, Polish (elementary)	